Naive Bayes Foundations: Conditional Probability

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Conditional probability.

1. Overview

Key point: The probability of A given B, P(A | B), is the probability of A once we know B has happened. P(A | B) equals P(A ∩ B) / P(B).

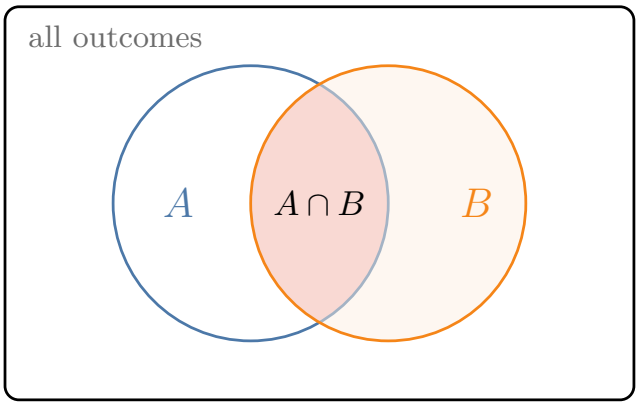
Later, these ideas build the Naive Bayes (G-1297) classifier, a very fast classification algorithm that works well on text, famously for document classification and spam filtering (scikit-learn user guide §1.9). It rests on a few ideas from probability: conditional probability, independent events, and **Bayes' theorem** (G-269). This Note covers the first.

Conditional probability (G-444) answers questions of the form "how likely is A, now that we know B is true?". Conditional probability is used throughout probability and machine learning, and Bayes' theorem is built directly on it.

2. The definition

Key point: P(A | B) = P(A ∩ B) / P(B), as long as P(B) is not zero.

An everyday picture: a friend draws a card from a deck and says "it is red". We stop thinking about all 52 cards and look only at the 26 red ones. The chance that the card is a heart is now 13 out of 26, one half, instead of one quarter.



$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

once B is known to have happened, B becomes the whole world: what share of B is also A?

Figure 1: Conditional probability as a share of B

The idea (Figure 1): once we know B has happened, every outcome outside B is ruled out. B becomes the whole world, and the question is what share of it also belongs to A.

Written as a formula, for two events (G-717; things that can happen, such as "the card is a heart") A and B, as long as $P(B)$ is not zero:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

- $P(A \mid B)$ is read "the probability of A given B".
- $A \cap B$ (A **intersection**, G-967) means "both A and B happen".
- The top is the part of B that also belongs to A; the bottom is all of B. For the card:

$$P(\text{heart and red}) = \frac{13}{52}$$

$$P(\text{red}) = \frac{26}{52}$$

$$P(\text{heart} \mid \text{red}) = \frac{13/52}{26/52}$$

$$= \frac{13}{26} = \frac{1}{2}$$

The condition that $P(B)$ is not zero is there because $P(B)$ sits in the bottom of the fraction, and division by zero has no value. An event with probability 0 never happens, so there is no "world of B" to look inside.

3. An example with two dice

> **Key point:** Rolling two dice has 36 equally likely outcomes. Conditioning on "sum $\leq$ 10" shrinks the space to 33 outcomes.

Roll two dice together. Each outcome is a pair, (die 1, die 2): (1, 1), (1, 2), ..., (6, 6). There are 36 outcomes, all equally likely:

$$6 \times 6 = 36$$

This set of all outcomes is the sample space (G-1729).

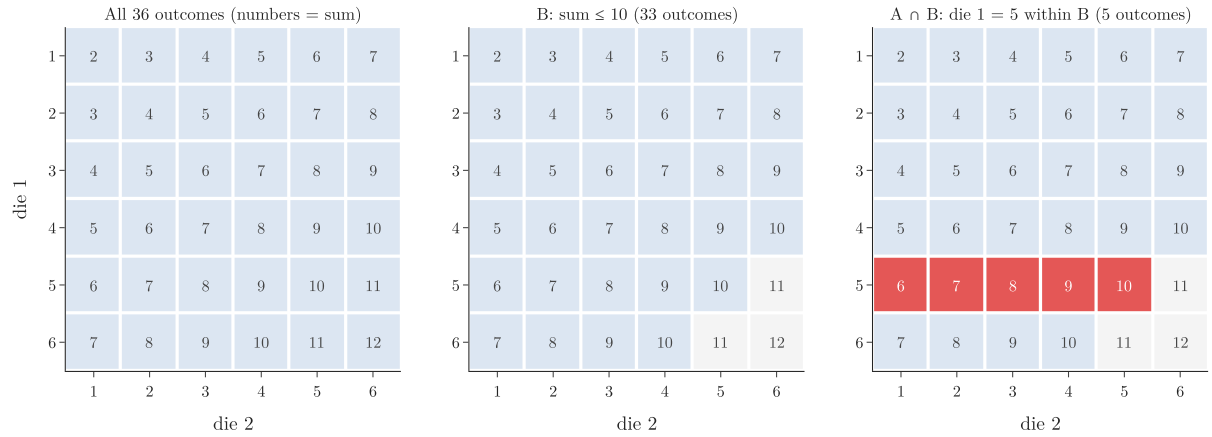


Figure 2: The 36 outcomes, the event B, and A ∩ B

Figure 2 lays the sample space out as a grid, with the sum of the two dice in each cell.

3.1 Two ordinary probabilities

> **Key point:** P(die 1 = 5) = 6/36; P(sum ≤ 10) = 33/36.

With equally likely outcomes, a probability is the number of favourable outcomes divided by the total.

- **A: die 1 shows 5.** Event A is the whole fifth row of the grid: 6 outcomes.

$$P(A) = \frac{6}{36} = \frac{1}{6}$$

- **B: the sum is at most 10.** Only three outcomes break it: (5, 6), (6, 5) and (6, 6), with sums 11, 11 and 12. So 33 outcomes are left (Figure 2, middle).

$$P(B) = \frac{33}{36} = \frac{11}{12}$$

3.2 A conditional probability by counting

> **Key point:** Count only inside B: 5 of its 33 outcomes have die 1 = 5, so P(A | B) = 5/33.

Now ask: **what is the probability that die 1 shows 5, given that the sum is at most 10?**

Since B is known to have happened, only its 33 outcomes are possible. These 33 outcomes form the **reduced sample space** (G-1649). Inside it, die 1 shows 5 in the outcomes (5, 1),

(5, 2), (5, 3), (5, 4) and (5, 5); the sixth, (5, 6), has sum 11 and is ruled out (Figure 2, right). So

$$P(A \mid B) = \frac{5}{33} \approx 0.152$$

Knowing B has lowered the chance of A slightly, because B ruled out one of A's outcomes. Without B, the chance was

$$P(A) = \frac{6}{36} \approx 0.167$$

Figure 3 shows the shrinking as motion, after the geometric picture of Bayes' rule by Sanderson (3Blue1Brown). Watch the three cells outside B disappear; the 33 cells left are then laid out again to fill the same area as the original 36, because B is now the whole world. A's 5 cells (red) take 5/33 of it. The last scene swaps the roles, as in section 4.

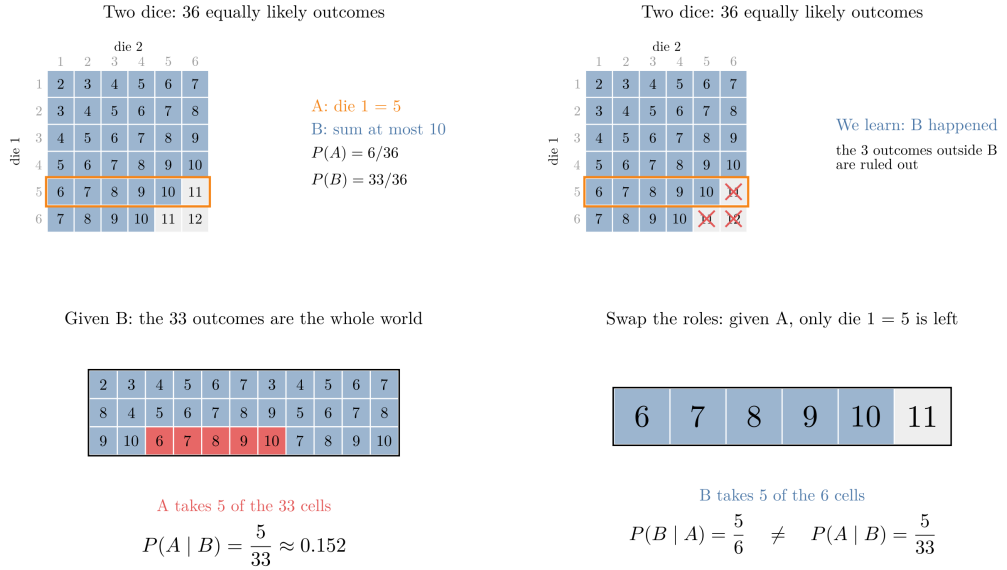


Figure 3: Conditioning shrinks the sample space: the 3 outcomes outside B are ruled out, the 33 left become the whole world and A covers 5 of them; given A instead, B covers 5 of A's 6 outcomes

3.3 The same answer from the formula

> **Key point:** P(A ∩ B) = 5/36 and P(B) = 33/36; dividing gives 5/33.

Using the formula instead, with the full **sample space** (G-1729) of 36:

- $P(A \cap B)$: die 1 is 5 **and** the sum is at most 10. These are the same 5 outcomes.

$$P(A \cap B) = \frac{5}{36}$$

- The sum is at most 10 in 33 outcomes.

$$P(B) = \frac{33}{36}$$

$$P(A \mid B) = \frac{5/36}{33/36} = \frac{5}{33}$$

The 36s cancel, leaving exactly the count in the **reduced sample space** (G-1649). The formula and the counting say the same thing.

> **Python:** Counting all 36 outcomes.
> ```
> from fractions import Fraction
> from itertools import product
>
> outcomes = list(product(range(1, 7), repeat=2))
> B = [o for o in outcomes if sum(o) <= 10]          # 33 outcomes
> A_and_B = [o for o in B if o[0] == 5]              # 5 outcomes
> Fraction(len(A_and_B), len(B))                     # 5/33
> ```
> A simulation of a million rolls gives 0.1512, close to the exact 0.1515, which is 5 divided by 33.

4. P(A | B) is not P(B | A)

> **Key point:** Swapping the roles changes the answer: P(sum $\leq$ 10 | die 1 = 5) = 5/6, not 5/33.

The order matters. Given that die 1 shows 5, the reduced space is its 6 outcomes, of which 5 have a sum of at most 10:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{5/36}{6/36} = \frac{5}{6}$$

So the two conditional probabilities differ (Figure 3, last scene):

$$P(B \mid A) = \frac{5}{6}$$

$$P(A \mid B) = \frac{5}{33}$$

Figure 4 sets the two side by side on the grid. The overlap is the same 5 red cells in both panels; what changes is the world we divide by: B's 33 cells on the left, A's 6 cells on the right.

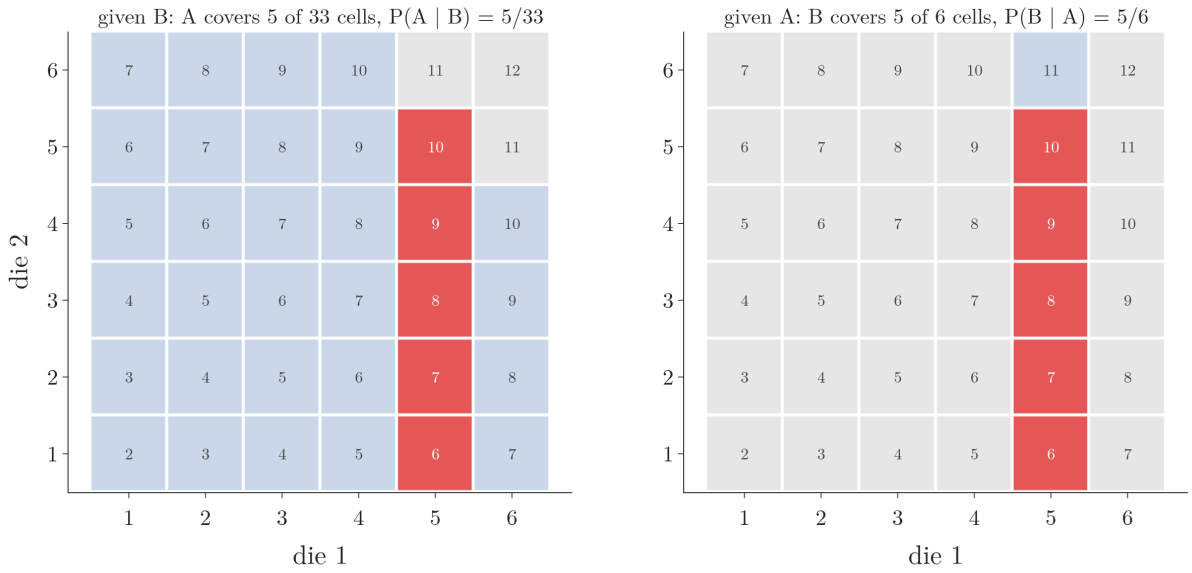


Figure 4: The same overlap, two different worlds. Left: given B (the 33 cells with sum at most 10), A covers 5 cells, so P(A | B) = 5/33. Right: given A (die 1 shows 5), B covers 5 of its 6 cells, so P(B | A) = 5/6. Grey cells are ruled out.

Confusing the two is a classic mistake. **Bayes' theorem** (G-269) is exactly the rule that converts one into the other.

5. The multiplication rule

> **Key point:** P(A ∩ B) = P(A | B) × P(B) = P(B | A) × P(A). The chance that both happen can be built from either side.

Multiply both sides of the definition by $P(B)$:

$$P(A \cap B) = P(A \mid B) \times P(B)$$

In words: for both to happen, B must happen, and then A must happen given B. This is the **multiplication rule** (G-2212). The same event can be built from the other side, starting with A:

$$P(A \cap B) = P(B \mid A) \times P(A)$$

An everyday example. Rahul likes bagels and pizza.

- A: he eats a bagel for breakfast. $P(A) = 0.6$.
- B: he eats pizza for lunch. $P(B) = 0.5$.
- On the days he has pizza for lunch, he had a bagel for breakfast 7 times out of 10: $P(A \mid B) = 0.7$.

What is $P(B \mid A)$, the chance of pizza for lunch given a bagel for breakfast?

1. Build "both" from the B side:

$$P(A \cap B) = 0.7 \times 0.5 = 0.35$$

2. The A side must give the same number:

$$P(B \mid A) \times 0.6 = 0.35$$

3. Divide:

$$P(B \mid A) = \frac{0.35}{0.6} \approx 0.58$$

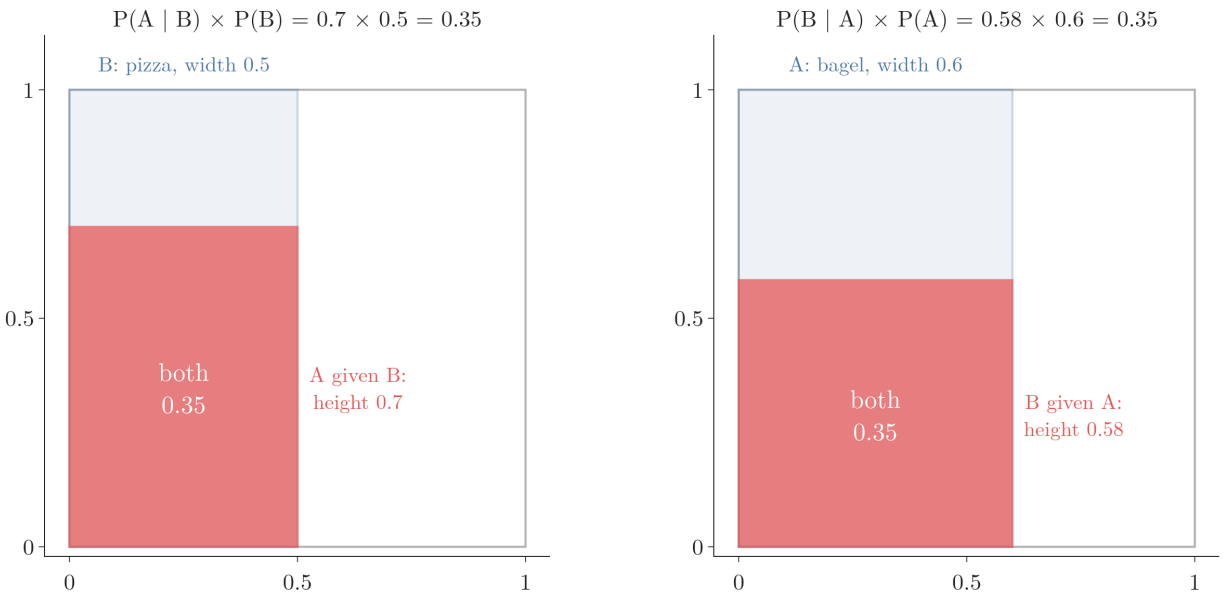


Figure 5: The multiplication rule as an area. Left: the B strip is 0.5 wide and A fills 0.7 of its height. Right: the A strip is 0.6 wide and B fills 0.58 of its height. The red area, the chance of both, is 0.35 in each.

Figure 5 shows the rule as an area. The whole square is every day, with area 1. On the left, the strip for B is 0.5 wide and A fills 0.7 of it, so the red rectangle has area 0.35. On the right, the strip for A is 0.6 wide and B fills 0.58 of it: the same area, 0.35.

Two things to take from the example:

- **The order matters again:** $P(A \mid B) = 0.7$ but $P(B \mid A) = 0.58$, as in Section 4.
- **The events affect each other:** $P(A \mid B) = 0.7$ is not $P(A) = 0.6$, so knowing B changes the chance of A. Such events are dependent events (G-590). When knowing B changes nothing, the events are independent (the probability of one does not change when the other is known).

Setting the two forms of the rule equal and dividing, as in steps 2 and 3, is **Bayes' theorem** (G-269).

6. Summary

- **Conditional probability** (G-444): $P(A \mid B) = P(A \cap B)/P(B)$, because once B has happened, B is the whole world and we ask what share of it is also A.
- Knowing B shrinks the **sample space** (G-1729) to B; count A's share inside it. The formula gives the same answer, because the full total cancels.
- **Multiplication rule** (G-2212): $P(A \cap B) = P(A \mid B) \times P(B) = P(B \mid A) \times P(A)$, so the chance of both can be built from either side; setting the two sides equal gives **Bayes' theorem** (G-269).
- Two dice: the chance of a 5 on die 1 given a sum of at most 10 is 5 in 33; the chance of a sum of at most 10 given a 5 on die 1 is 5 in 6. Swapping the condition changes the world we divide by, so $P(A \mid B)$ is not $P(B \mid A)$; Bayes' theorem converts one into the other.
- Conditional probability answers "how likely is A, now that we know B?", and Bayes' theorem, and through it **Naive Bayes** (G-1297), is built directly on it.

7. Sources

Built from

- CampusX, "Naive Bayes Classifier | Part 1 | Conditional Probability", YouTube, https://www.youtube.com/watch?v=Ty7knppVo9E
- Khan Academy, "Calculating conditional probability | Probability and Statistics", YouTube, https://www.youtube.com/watch?v=6xPkG2pA-TU (the breakfast-and-lunch example and the multiplication rule read both ways)
- **Sanderson, G. (3Blue1Brown):** "Bayes' theorem", 3blue1brown.com/lessons/bayes-theorem. The picture of conditioning as shrinking the space of possibilities; Figure 3 recreates it with our own dice example.

Other references

- **scikit-learn user guide:** Section 1.9, "Naive Bayes", scikit-learn 1.9.

8. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Conditional probability (G-444)	The probability of an event given that another event has happened: $P(A \mid B)$.
Intersection (A ∩ B) (G-967)	The set of elements that are in both A and B, such as the points inside both of two half-planes; for events, the event that both A and B happen.

Term	Meaning
Reduced sample space (G-1649)	The outcomes that remain possible once the condition is known.
Multiplication rule (G-2212)	The chance that both A and B happen is the chance of B times the chance of A given B: $P(A \cap B) = P(A \mid B) \times P(B)$.
Naive Bayes classifier (Naive Bayes) (G-1297)	A classifier that applies Bayes' theorem with the assumption that inputs are independent within each class.
Bayes' theorem (G-269)	The rule that reverses a conditional probability: from $P(B \mid A)$ it gives $P(A \mid B)$, so a belief about A can be updated after seeing B; $P(A \mid B) = P(B \mid A)P(A)/P(B)$.
Event (G-717)	A set of outcomes, such as "the sum is at most 10".
Sample space (G-1729)	The set of all possible outcomes of an experiment, such as $\{1, \ldots, 6\}$ for a die; every event is a subset of it, so probabilities are worked out inside it.
Dependent events (G-590)	Events that are not independent: knowing one changes the probability of the other.